

Numerical Summaries

A numerical summary for a set of data is referred to as a statistic if the data set is a sample and a parameter if the data set is the entire population.

Numerical summaries are categorized as measures of location and measures of spread.

Measures of location can further be classified into measures of central tendency and measures of relative positioning (quantiles).

Measures of Location

Measures of Central Tendency

A Measures of Central Tendency of a set of numbers is a value which best represents it.

There are three different types of Central Tendencies namely the mean, median and mode.

Each has advantages and disadvantages depending on the data and intended purpose.

Arithmetic Mean The arithmetic mean of a set of values $x_1, x_2, \dots, x_n$ denoted by $\bar{x}$ if the data set is a sample is found by dividing the sum of the set of numbers with the actual number of values

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

Example

Find the mean of 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10.

Solution

The sum of values:

$$\sum_{i=1}^n x_i = 55$$

Thus the mean is:

$$\bar{x} = \frac{55}{10} = 5.5$$

In R the mean is obtained as:

```
# reading in the values
x <- c(1,2,3,4,5,6,7,8,9,10)

# obtaining the mean
mean(x)
```

```
[1] 5.5
```

Properties of Arithmetic Mean

1. The algebraic sum of the deviations of a set of numbers from their arithmetic mean is zero, that is $\sum_{i=1}^n (x_i - \bar{x}) = 0$
2. $\sum_{i=1}^n (x_i - a)^2$ is minimum if and only if $a = \bar{x}$
3. If n_1 numbers have mean $\bar{x}_1$, n_2 numbers have mean $\bar{x}_2, \dots, n_k$ numbers have mean $\bar{x}_k$ then the mean of all the numbers called the combined mean is given by:

$$\bar{x}_c = \frac{n_1\bar{x}_1 + \dots + n_k\bar{x}_k}{n_1 + n_2 + \dots + n_k} = \frac{\sum n_i\bar{x}_i}{\sum n_i}$$

Median

It is the value below which and above which half of the observations fall when ranked in order of size. The position of the median term is given by:

$$\text{Position} = \frac{n + 1}{2}$$

NB: If n is even, we average the middle 2 terms.

Grouped Data

For grouped data, the median is estimated using the formula:

$$\text{Median} = \text{LCB} + \left(\frac{\frac{N}{2} - \text{CF}_a}{f} \right) \times i$$

Where:

- **LCB:** Lower class boundary of the median class.
- **f:** Frequency of the median class.
- **i:** Class interval size.
- **CF_a:** Cumulative frequency of the class preceding (above) the median class.

Remark: The disadvantage of the median is that it is not sensitive to changes in the data.

Mode

It is the value occurring most frequently in a data set. If each observation occurs the same number of times, then there is no mode. When 2 or more observations occur most frequently in a dataset, the data is said to be multimodal.

For ungrouped data, it is very easy to pick out the mode. However, if the data is grouped, the mode is estimated using the formula:

$$\text{Mode} = \text{LCB} + \left(\frac{f - f_a}{(f - f_a) + (f - f_b)} \right) \times i$$

Where:

- **LCB:** Lower class boundary of the modal class.
- **f:** Frequency of the modal class.
- **i:** Class interval size.
- f_a : Frequency of the class above (preceding) the modal class.
- f_b : Frequency of the class below (following) the modal class.

Remark: The Empirical Relation between the Mean, Median, and Mode

$$\text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median}) \quad \text{or} \quad \text{Mode} \approx 3 \text{Median} - 2 \text{Mean}$$

The above relation is true for unimodal frequency curves which are asymmetrical.

Examples

Example 1

Find the median and mode for the data: 19, 13, 18, 14, 12, 25, 11, 17, 16, 23, 19.

Solution

Sorted data: 11, 12, 13, 14, 16, 17, 18, 19, 19, 23, 25 ($n = 11$).

$$\text{Position of Median} = \frac{11 + 1}{2} = 6\text{th item} \implies \text{Median} = 17$$

Mode = 19 since it appears most frequently in this data set compared to other observations.

Example 2

Find the median and mode of the data: 2, 4, 8, 7, 9, 4, 6, 10, 8, and 5.

Solution

Array (Sorted): 2, 4, 4, 5, 6, 7, 8, 8, 9, 10 ($n = 10$). \ Mode = 4 and 8 (i.e., the data is bimodal).

$$\text{Position} = \frac{10 + 1}{2} = 5.5\text{th item} \implies \text{Median} = \frac{6 + 7}{2} = 6.5$$

Example 3

Estimate the mean, median, and mode for the following frequency distribution:

Class	5-9	10-14	15-19	20-24	25-29	30-34	35-39
Freq	5	12	32	40	16	9	6

Class Boundaries	Midpoints (x)	Frequency (f)	xf	CF
4.5 – 9.5	7	5	35	5
9.5 – 14.5	12	12	144	17
14.5 – 19.5	17	32	544	49
19.5 – 24.5	22	40	880	89
24.5 – 29.5	27	16	432	105
29.5 – 34.5	32	9	288	114
34.5 – 39.5	37	6	222	120
Total		$N = 120$	$\sum xf = 2565$	

Solution

Mean:

$$\text{Mean} = \frac{\sum xf}{N} = \frac{2565}{120} = 21.375$$

Median: \ The median position is $\frac{120}{2} = 60$. The median class is 19.5 – 24.5. Thus:

$$\text{Median} = 19.5 + \left(\frac{60 - 49}{40} \right) \times 5 = 19.5 + 1.375 = 20.875$$

Mode: \ The modal class (class with the highest frequency) is 19.5 – 24.5. Therefore:

$$\text{Mode} = 19.5 + \left(\frac{40 - 32}{(40 - 32) + (40 - 16)} \right) \times 5 = 19.5 + \left(\frac{8}{8 + 24} \right) \times 5 = 20.75$$

Other Types of Means

These will include weighted, harmonic, and geometric means.

The Weighted Arithmetic Mean

The weighted arithmetic mean of a set of n numbers $x_1, x_2, \dots, x_n$ having corresponding weights $w_1, w_2, \dots, w_n$ is defined as:

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

Example 1

Consider the following table with marks obtained by two students James (mark x) and John (mark y). The weights are to be used in determining who joins the engineering course whose requirement is a weighted mean of 58% on the four subjects below:

Working out the products of the marks and the weights, we get:

Subject	Maths	English	History	Physics	Total
Mark x	25	87	83	30	225
Mark y	70	45	35	75	225
Weight (w)	3.6	2.3	1.5	2.6	10

Subject	Maths	English	History	Physics	Total
wx	90.0	200.1	124.5	78.0	492.6
wy	252.0	103.5	52.5	195.0	603.0

Now computing the weighted averages:

$$\bar{x}_{\text{James}} = \frac{492.6}{10} = 49.26\%$$

$$\bar{y}_{\text{John}} = \frac{603.0}{10} = 60.3\%$$

Clearly, John qualifies but James does not.

Example 2

If a final examination is weighted 4 times as much as a quiz, a midterm examination 3 times as much as a quiz, and a student has a final examination grade of 80, a midterm examination grade of 95, and quiz grades of 90, 65, and 70, the mean grade is:

$$\bar{x}_w = \frac{(4 \times 80) + (3 \times 95) + (1 \times 90) + (1 \times 65) + (1 \times 70)}{4 + 3 + 1 + 1 + 1} = \frac{320 + 285 + 90 + 65 + 70}{10} = \frac{830}{10} = 83.0$$

Question 1 (Tycoon Salaries)

A tycoon has 3 house girls who he pays Ksh 4,000 each per month, 2 watchmen who he pays Ksh 5,000 each, and some garden men who receive Ksh 7,000 each. If he pays out an average of Ksh 5,700 per month to these people, find the number of garden men.

Solution: Let g be the number of garden men. The total number of workers is $3 + 2 + g = 5 + g$.

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

$$5700 = \frac{(3 \times 4000) + (2 \times 5000) + (g \times 7000)}{5 + g}$$

$$5700(5 + g) = 12000 + 10000 + 7000g$$

$$28500 + 5700g = 22000 + 7000g$$

$$28500 - 22000 = 7000g - 5700g$$

$$6500 = 1300g$$

$$g = \frac{6500}{1300} = 5$$

There are **5** garden men.

Question 2 (Computer Course Grades)

A student's grades in laboratory, lecture, and recitation parts of a computer course were 71, 78, and 89, respectively.

- If the weights accorded these grades are 2, 4, and 5, respectively, what is the average grade?
- What is the average grade if equal weights are used?

Solution:

- Using the given weights:

$$\bar{x}_w = \frac{(2 \times 71) + (4 \times 78) + (5 \times 89)}{2 + 4 + 5} = \frac{142 + 312 + 445}{11} = \frac{899}{11} \approx 81.73$$

- Using equal weights (simple arithmetic mean):

$$\bar{x} = \frac{71 + 78 + 89}{3} = \frac{238}{3} \approx 79.33$$

The Geometric Mean

The geometric mean is well suited for data measured as ratios, proportions, or percentages. For this type of data, it is more reliable than the arithmetic mean.

Definition: The geometric mean (G) for a set of observations $x_1, x_2, \dots, x_n$ is given by:

$$G = \sqrt[n]{x_1 \times x_2 \times \dots \times x_n} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}}$$

For grouped data, one has $G = \sqrt[N]{x_1^{f_1} \times x_2^{f_2} \times \dots \times x_k^{f_k}}$ where $N = \sum f_i$.

If n is too large, calculating the n -th root can be a challenge. In this case, the geometric mean is computed using logarithms:

$$\log G = \frac{1}{n} \sum \log x_i \quad \text{or for grouped data:} \quad \log G = \frac{1}{N} \sum f_i \log x_i$$

Taking the antilogarithm yields G , the geometric mean.

Example

Yearly percentage growth of urban population in Kenya from 1965 to 1970 is given below:

Year	1965	1966	1967	1968	1969	1970
% Increase	8.35	17.91	33.08	40.32	35.80	47.67

Obtain the average percentage growth rate of urban population from 1965 to 1970 in Kenya.

Solution:

$$\begin{aligned}\log G &= \frac{\log(8.35) + \log(17.91) + \log(33.08) + \log(40.32) + \log(35.80) + \log(47.67)}{6} \\ &= \frac{0.9217 + 1.2531 + 1.5196 + 1.6055 + 1.5539 + 1.6782}{6} = \frac{8.5320}{6} = 1.4220 \\ G &= \text{antilog}(1.4220) \approx 26.42\%\end{aligned}$$

Question 3 (Company Profits)

Below are the yearly percentage profits made by a company in six successive years: 52.22, 46.59, 21.36, 30.17, 22.87, 17.77. Determine the mean percentage profits made by the company.

Solution:

$$\begin{aligned}\log G &= \frac{\log(52.22) + \log(46.59) + \log(21.36) + \log(30.17) + \log(22.87) + \log(17.77)}{6} \\ &= \frac{1.7178 + 1.6683 + 1.3296 + 1.4796 + 1.3593 + 1.2497}{6} = \frac{8.8043}{6} \approx 1.4674 \\ G &= \text{antilog}(1.4674) \approx 29.34\%\end{aligned}$$

The Harmonic Mean

This type of mean is suitable for data pertaining to speed, rates, and time.

Definition: The Harmonic mean (H) for a set of observations $x_1, x_2, \dots, x_n$ is given by:

$$H = \frac{n}{\sum \frac{1}{x_i}}$$

For grouped data, we have $H = \frac{N}{\sum \frac{f_i}{x_i}}$ where $N = \sum f_i$. **Note:** This type of mean cannot be used when any observation x_i is zero.

Example

A production machine is programmed to run for four hours only per day. In the first hour, its output is 40 units/h, in the second hour its output is 60 units/h, in the third hour the output is 70 units/h while in the fourth hour the output is 30 units/h. Determine the average output per hour of the machine.

Solution: Using the Harmonic mean:

$$H = \frac{4}{\frac{1}{40} + \frac{1}{60} + \frac{1}{70} + \frac{1}{30}} = \frac{4}{0.0250 + 0.0167 + 0.0143 + 0.0333} = \frac{4}{0.0893} \approx 44.79 \text{ units/h}$$

Example: Geometric and Harmonic Mean for Grouped Data

Find the harmonic and geometric mean of the frequency table below:

x	13	14	15	16	17
f	2	5	13	7	3

x	f	f/x	$\log x$	$f \log x$
13	2	0.1538	1.1139	2.2278
14	5	0.3571	1.1461	5.7305
15	13	0.8667	1.1761	15.2893
16	7	0.4375	1.2041	8.4287
17	3	0.1765	1.2304	3.6912
Total	$N = 30$	$\sum \frac{f}{x} = 1.9916$		$\sum f \log x = 35.3675$

Solution Setup Table: Harmonic Mean (H):

$$H = \frac{N}{\sum \frac{f_i}{x_i}} = \frac{30}{1.9916} \approx 15.063$$

Geometric Mean (G):

$$\log G = \frac{\sum f_i \log x_i}{N} = \frac{35.3675}{30} \approx 1.1789$$

$$G = \text{antilog}(1.1789) \approx 15.097$$

Properties of Measures of Location

- (i) They are affected by change of origin. Adding or subtracting a constant from each and every observation in a data set causes all the measures of location to shift by the same magnitude.
- (ii) They are affected by change of scale. Multiplying each and every observation in a data set by a constant value scales up all the measures of location by the same magnitude.

Measures of Spread/Dispersion

Spread is the degree of scatter or variation of the variable about the central value.

Examples of these measures includes: the range, Inter-Quartile range, Quartile Deviation also called semi Inter-Quartile range, Mean Absolute Deviation, Variance and standard deviation.

Inter-Quartile Range and Semi Inter-Quartile Range

The **Inter-Quartile Range (IQR)** is the difference between the upper quartile (Q_3) and the lower quartile (Q_1). Half of this difference is called the **Quartile Deviation** or the **Semi Inter-Quartile Range (SIQR)**:

$$IQR = Q_3 - Q_1$$

$$SIQR = \frac{Q_3 - Q_1}{2}$$